

Macrium Reflection Workstation

Backup & Restore

Better Semiconductors
every day



Document History

Version	Status	Date	Responsible	Reason for Change
1.0.0	Development	25.02.2025	Oliver Karger	

Table of Content

Document History	2
Table of Content.....	3
1. Installation	4
1.1. License Key	4
1.2. Installer.....	4
1.3. Setup	4
2. Application Overview	5
3. Creating a Backup.....	6
3.1. Incremental / Differential Backups.....	7
3.2. Automatic Backups	7
4. Restoring a Backup	8
5. Rescue Media creation.....	10
5.1. Advanced Options	10

1. Installation

1.1. License Key

The License Key für Macrium Reflect for customer Projects will be ordered by Parts Lists and delivered to the PC-owner by the Warehouse Staff.

1.2. Installer

The Installer is located at [\\serv-aho-file\Daten\EDV\Software\Macrium Reflect](#)

Note: Please use the latest v8 Installer !

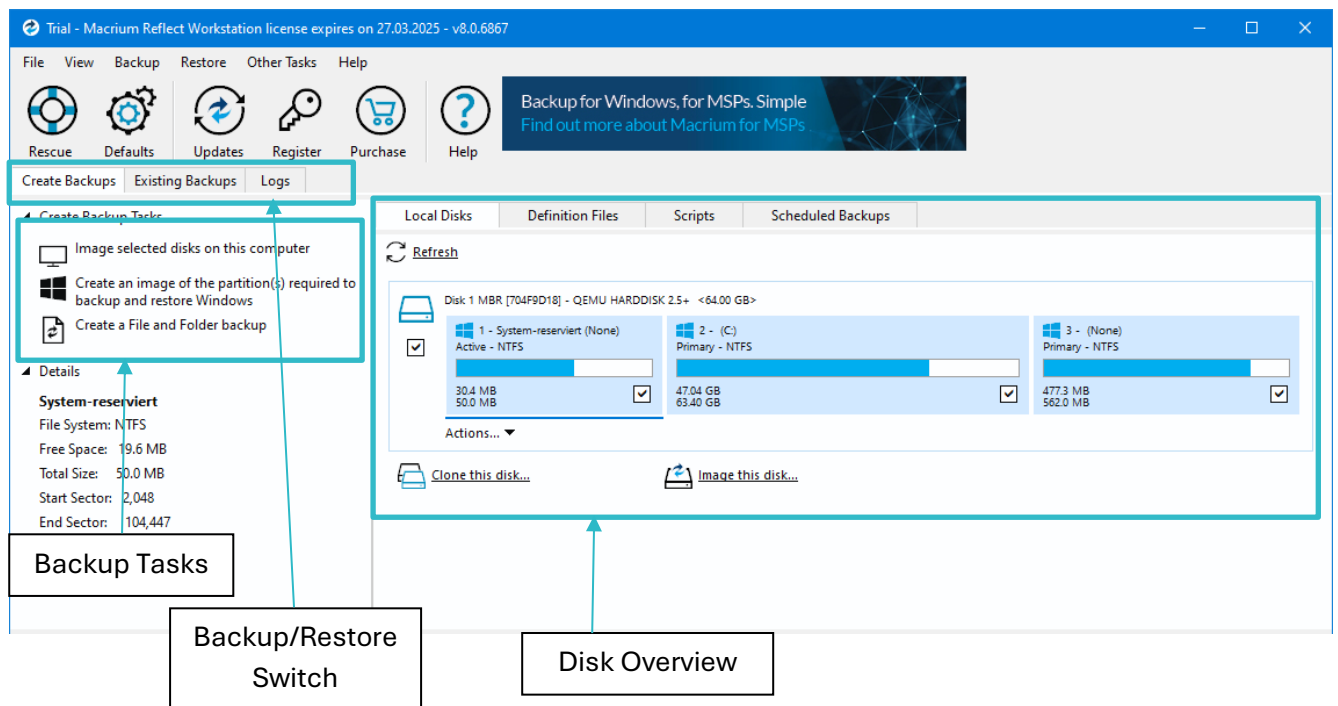
It's also recommended to copy to installer locally for faster Setup

1.3. Setup

1. Run the Installer „v8.<version>_reflect_wkstn_setup_x64.exe“
2. Click „Next“ and accept the EULA.
3. Uncheck „30 Day Trial“ and enter the License Key
4. Change settings for “Customized Installation”
 - Uncheck “CBT” (This would increase Backup Speed but at the cost of overall System Performance)
 - Uncheck “Image Guardian” (This encrypts Backups but requires a completely separated Drive that can after that only be used with Macrium)
 - Uncheck “ViBoot” (This allows to boot Macrium Backups as Virtual Machines with Hyper-V. That is not required on Tool Computers)
 - Check “Desktop Shortcut”
 - Leave the Path to the default Value
5. Finish the Installation by clicking “Next” and then “Install”

After the Setup is completed, restart the Computer.

2. Application Overview



Note: A Trial License was used for this Documentation – therefor certain Buttons like “Register” and “Purchase” will be missing if the Software has been activated with a proper License

3. Creating a Backup

Create an image of the partition(s) required to backup and restore Windows

1

Disk Image

Select Source Drive(s) & Partition(s)

Source

Disk	Partition	File System	Size	Selected
Disk 1 (MBR)	1 - System-reserviert (None)	Active - NTFS	39.4 MB	☒
Disk 2 (C:)	2 - (C:)	Primary - NTFS	47.04 GB	☒
Disk 3 (None)	3 - (None)	Primary - NTFS	477.3 MB	☒

Total Selected: 47.54 GB

Destination

☒ Folder

Alternative locations

☐ CD/DVD Burner

Backup File Name

Type '{' for parameters (IMAGEID)-00-00.nvimg

Advanced Options

Help < Back Next > Cancel Finish

2

1. Look for the “C” Drive and select the entire Row. That’s the Windows Installation
2. Look for the “M” Drive, if its not already covered by Step (1), select this one too

Default Backup Path is E:\!Backup or D:\!Backup
Ideally, created a Subfolder with Name Pattern “<Date> <Creator>”
E. G. 2025-02-25 OK

3

Disk Image

Edit the Plan for this Backup

1. Select a Template for your Backup Plan

None

2. Add/Edit Schedules

Backup Type	Schedule

+ Add Schedule Edit Schedule - Delete Schedule

3. Define Retention Rules

Apply retention rules to matching backup sets in the target folder

Retention Rule	Keep	Backups
☒ Full	12	Backups
☐ Differential	4	Backups
☐ Incremental	10	Backups

Create a Synthetic Full if possible

☐ Run the purge before backup.

☐ Purge the oldest backup set(s) if less than 5 GB on the target volume (minimum 1GB)

Advanced Options

Help < Back Next > Cancel Finish

Uncheck all Settings.

These are settings specific to Incremental and Differential Backups.

We currently don’t offer this as a “Service” to our Customers – therefore we disable these Settings for “our” Backups.

On the next Page, you’ll see a Overview of the Backup Settings you did. Please check here that all Drives you want to be included in the Backup are listed and that the correct Destination Path is set

Backup Save Options

What do you want to do now?

☒ Run this backup now

☒ Save as a Backup Definition File

The saved Backup Definition enables you to quickly rerun this configuration.

Enter a name for this Backup Definition.

My Backup

C:\Users\ClientsAdmin\Documents\Reflect\My Backup.xml

Help OK Cancel

4

Check the correct Boxes and click “OK” to start the Backup

3.1. Incremental / Differential Backups

1. Select a Template for your Backup Plan

None

None
Create your own schedules by selecting this option.

Grandfather, Father, Son.
Daily Incremental ("Son"), weekly Differential ("Father"), and monthly Full ("Grandfather") backups.

Differential Backup Set
A Full backup is created periodically followed by daily Differential backups.

Incremental Backup Set
A Full backup is created periodically followed by daily Incremental backups.

Intra-Daily Backup Set

Can be Configured on Step (3) of the Backup Procedure

3.2. Automatic Backups

1. Select a Template for your Backup

None

2. Add/Edit Schedules

Backup Type	Schedule

+ Add Schedule

Full

Differential

Incremental

Schedules for Full, Differential and Incremental Backups can be configured here (Also Step (3))

Local Disks | Definition Files | Scripts | Scheduled Backups | Task Status

Open | Add | Edit | Delete | Run | Schedule | Shortcut | Advanced | Filter List: All

File Name	Path
Docu.xml	C:\Users\oliver.kargen\Desktop\

Saved Backups can be ran from this Section in "Create Backup" – "Definition Files"

4. Restoring a Backup

For restoring a Backup, it is necessary that the “new Hardware” is not significantly different to the old one. Windows has Issues with Changing Hardware while Windows is installed due to Drivers etc. While it can be restored to different Hardware, it is not guaranteed that Windows will boot.

1

2

3

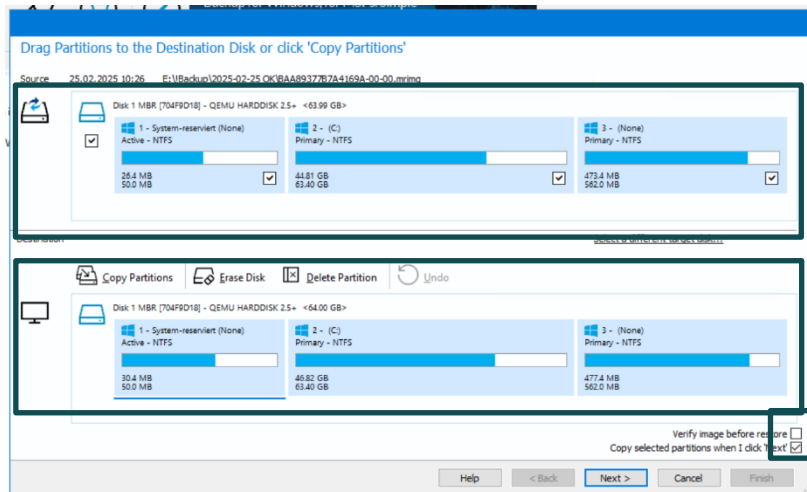
4

Selected Folder
E:\Backup\2025-02-25 OK

Path	Definition	ImageID	Type	Date	Size	Comment
E:\Backup\2025-02-25 OK\						
BAA89377B7A4169A-00-00.mrimg		BAA89377B7A4169A	Full	25.02.2025 10:26	25.10 GB	

Make sure the correct File is selected here!
If so, press “Verify” and then “Restore”

Ensure that both Source and Target Drives are correct and all Drives you want to recovery are listed in “Source”



This will overwrite all Data and Partitions on the Target Drive!

5

After this, you'll be presented with a similar “Overview” to when you created the Backup. Always Double Check the Information here to ensure a correct Restore!

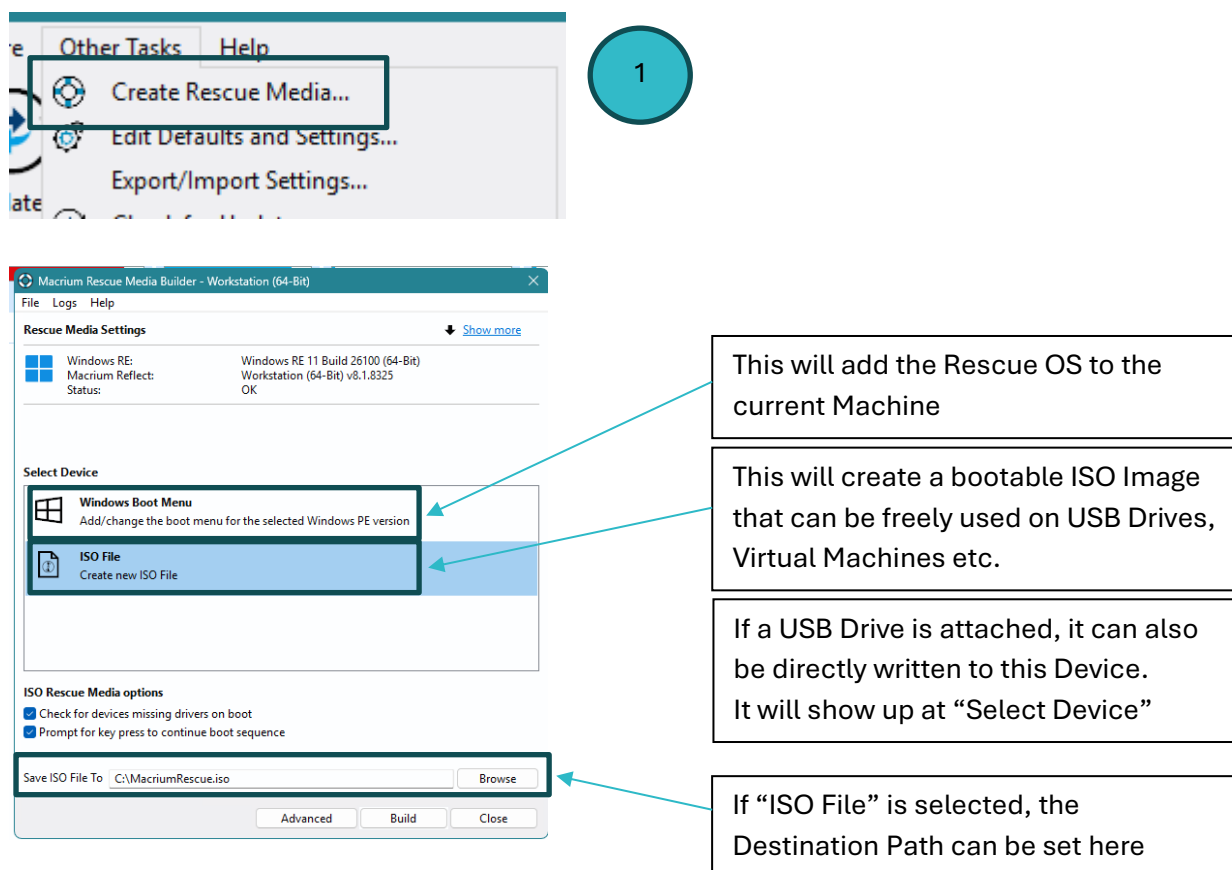
5. Rescue Media creation

Due to how Macrium Works, it is recommended that a Rescue Media is created for each Device Type. Ideally, a specialized Rescue Media for each Computer.

A Rescue Media is a bootable Drive with a temporary Windows Operating System with Macrium pre-Installed that can be used if the Host Operating System is inaccessible

Windows requires certain Drives to be present, if these are missing – Windows will not boot (“System Device Missing” Error)

When creating this Rescue Media for a specific Device, all required Drivers will automatically be copied and setup in the created Media.



The screenshot shows the Macrium Rescue Media Builder - Workstation (64-Bit) window. A red circle with the number '1' highlights the 'Create Rescue Media...' button in the top menu. The main window is divided into sections: 'Rescue Media Settings' (showing Windows RE and Macrium Reflect status), 'Select Device' (with 'Windows Boot Menu' and 'ISO File' options), 'ISO Rescue Media options' (with checkboxes for missing drivers and boot sequence), and a 'Save ISO File To' field with a 'Browse' button. Annotations with arrows point to these sections:

- Windows Boot Menu:** This will add the Rescue OS to the current Machine
- ISO File:** This will create a bootable ISO Image that can be freely used on USB Drives, Virtual Machines etc.
- ISO Rescue Media options:** If a USB Drive is attached, it can also be directly written to this Device. It will show up at “Select Device”
- Save ISO File To:** If “ISO File” is selected, the Destination Path can be set here

5.1. Advanced Options

In the “Advanced Options” Section (opened by clicking “Advanced” Button), there are several different Options that can be configured (only relevant ones listed)

- Devices & Drivers
 - All Devices detected whose Drivers can be copied to the Rescue Media